

Green Action Plan



**Department of
Forests and Wildlife**

Govt. of NCT of Delhi

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Green Action Plan (GAP)

1. Introduction:

The Department of Forests and Wildlife, Govt. of NCT of Delhi, being the Nodal Agency, undertakes the plantation drive under Green Action Plan in collaboration with more than 20 other Greening Agencies of the State. Additionally, free distribution of seedlings is also carried out by the nurseries of Department of Forests and Wildlife, Govt., of NCT of Delhi and Delhi Parks and Garden Society for encouraging general public towards greening of the city.

2. Implementing agencies:

Department of Forests and Wildlife, GNCTD , Delhi Development Authority (DDA), Municipal Corporations of Delhi (MCD), New Delhi Municipal Corporation (NDMC), Public Works Department, GNCTD (PWD), School Education, Directorate of Higher Education, National Thermal Power Corporation Limited (NTPC), Delhi State Industrial and Infrastructure Development Corporation Ltd. (DSIIDC), Delhi Jal Board (DJB), BSES Rajdhani, Central Public Works Department (CPWD), Northern Railways, Delhi Urban Shelter Improvement Board (DUSIB), Delhi Cantonment Board (DCB), Tata Power Delhi Distribution Limited (TPDDL), National Highways Authority of India (NHAI), Delhi Metro Rail Corporations Lmt. (DMRC), Irrigation and Flood Control Department, GNCTD (IF&CD), Department of Env. Horticulture, GNCTD, Delhi Transport Corporations (DTC), Delhi Parks and Gardens Society (DPGS).

3. Major initiatives for improving Green Cover of Delhi

3.1 Various plantation schemes to be implemented under GAP

3.1.1 Conventional Plantation: Planting evenly spaced saplings while ensuring their health and survival by watering, weeding, and application of manure is conventional plantation.

It has been recommended that minimum 60% of total saplings planted should be saplings of native species. Further it is to be ensured that minimum 25% seedlings of flowering species and 15% seedlings of fruit bearing species will be planted while undertaking plantation along roadside, in parks, gardens, city forests and residential areas. Suggestive list of native species for plantation is annexed herewith as **Annexure-I**. The suggestive list of fruiting species for plantation outside Forest Area is annexed herewith as **Annexure-II**. The suggestive list of flowering species for plantation in parks, gardens, city forests, residential areas and along roadside is annexed herewith as **Annexure-III**.

3.1.2 Avenue Plantation: Planting trees in the avenues, parks, gardens and city forests outside notified forest area of Delhi is of utmost importance due to the many benefits that plants provide. Thus, the Department of Forests and Wildlife, Government of Delhi has prepared a scheme of plantation for the city's public agencies. The dual purpose for selecting this species is greening and beautification.

3.1.3.1 The planting of trees and shrubs is categorized in four categories

- i. Tree plantation along roadways, turfing with grass and herbs;
- ii. Shrub plantation on medium strip, turfing with grass and herbs;
- iii. Block plantations in parks and gardens
- iv. Grid plantations in large patches of land

3.1.3.2 Selection of Species

Selection of trees, shrubs, herbs and grass species is based on following characteristics:

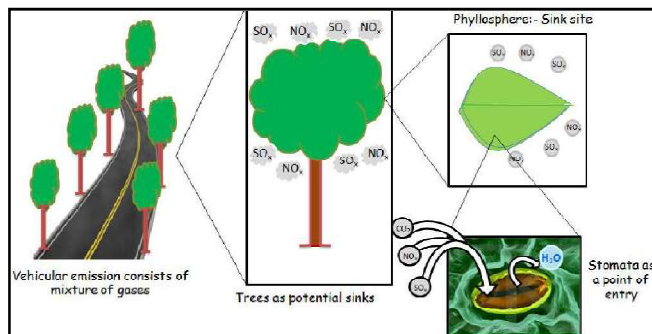
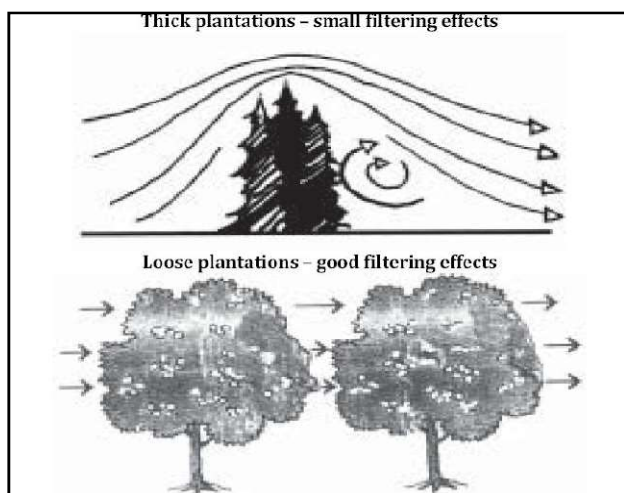
- i. Location of plantation site
- ii. Agro-climatic condition
- iii. Ecological importance
- iv. Presence, size and color of flowers
- v. Value for pollinators
- vi. Contribution towards pollution sequestration
- vii. Growth rate
- viii. Shape of the trunk
- ix. Foliage and branching pattern
- x. Shape and size of the canopy
- xi. Properties of nitrogen fixation
- xii. Soil-moisture conservation
- xiii. Socioeconomic importance
- xiv. Aftercare and maintenance requirement

A mix of showy species for beautification and local species commonly found in NCT of Delhi must be preferred.

Criteria for selection of species for roadside plantation¹

Agro-climatic Zone	Pollution Sequestration	Foliage Pattern	Growth Rate	Urban Beautification Value
As per the Agro-climatic map of ICAR, NCT of Delhi falls between the transition zone of upper and trans Gangetic Plains. (Figure-2)	Canopy Shape and Size: Dense foliage helps in better trapping of pollutants and also works as wind break. Openness in canopy allows better filtration capacity. Large canopy surface area works as better noise barrier. Leaf Area: Large and broad leaf size accelerates better pollution retention. Similarly leaves with complex shape and large circumference has higher dust capture capacity. Stomatal Index: High stomatal density, high index and large stomata give high transpiration rate thereby allowing higher sorption capacity. Tolerance: Selection of species with better tolerance capability for primary pollutants helps in maintaining the longevity of green corridor. (Figure-1)	Evergreen tree/shrub has higher pollution attenuation capacity as compared to deciduous or semi-deciduous tree/shrub.	Trees/shrubs with higher growth rate accelerate earlier canopy formation which is better for pollution sorption.	Shape of canopy. Presence, size, color and season of flowering. Ability to provide nectar and nesting space pollinators.

¹ Sharma, Anuj. "Plantation Species Matrix for Roadside Plantation". Report for National Green Highways Mission-NHAI, 2016.



Pollution Sequestration by Roadside Plantation

(Source: Sharma, Anuj. “Plantation Species Matrix for Roadside Plantation”. Report for National Green Highways Mission- NHAI, 2016)

In accordance to the guidelines (2009) of Indian Road Congress on landscaping and transplantation the space available in the ROW is one major guiding factor for different themes of landscaping applied in the entire corridor. And as one of the main objectives of roadside plantation is to provide shade, therefore, selection of plantation pattern for roadside plantation shall be designed as follows:

- i. The first row along the Highways will be of small to medium sized ornamental trees.
- ii. Subsequent rows, depending on the availability of width, will comprise of ornamental and/or shade bearing species, of more height than those in the first row.
- iii. All these trees to be capable of attenuating pollution.
- iv. Plantation of shrubs in the median.
- v. Turfing with herbaceous species as ground cover in the median.
- vi. Turfing with grass in the median.

3.1.3.3 Specifications for Road Plantation

As per “Guidelines on Landscaping and Transplantation”, Indian Road Congress, 2009 the technical specifications for roadside plantation are given below:

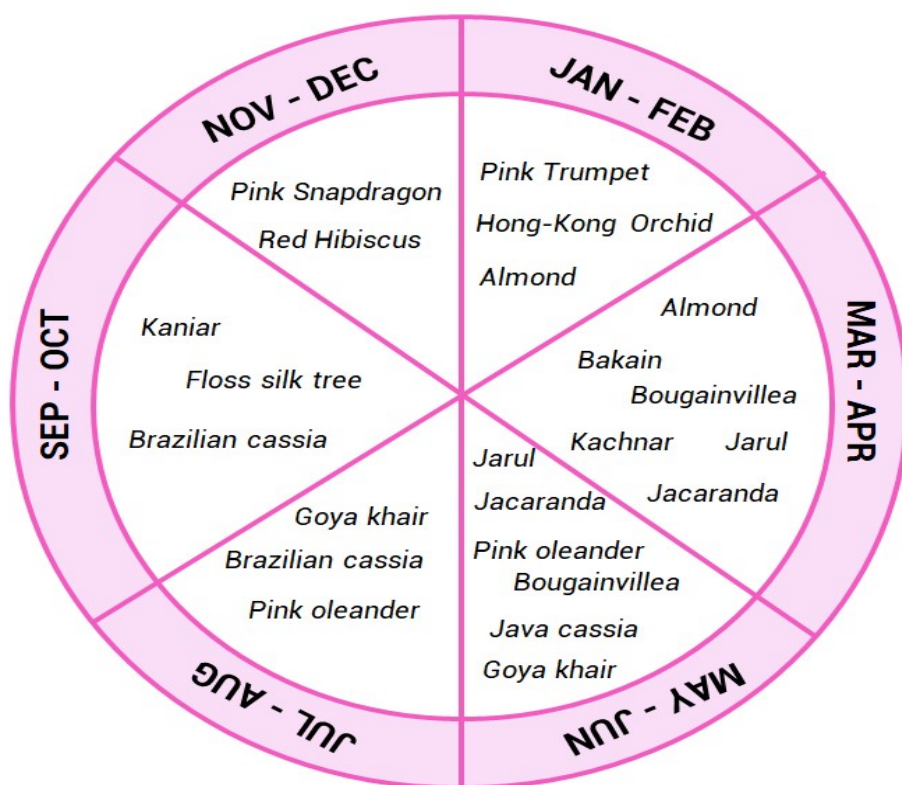
Specifications for ornamental plants (except last row)

1.	Spacing between plant to plant	3 m
2.	Spacing between rows	3 m
3.	Size of the pits	60x60x60 cm (in alkaline soils, kankar panes to be broken by augur. In waterlogged

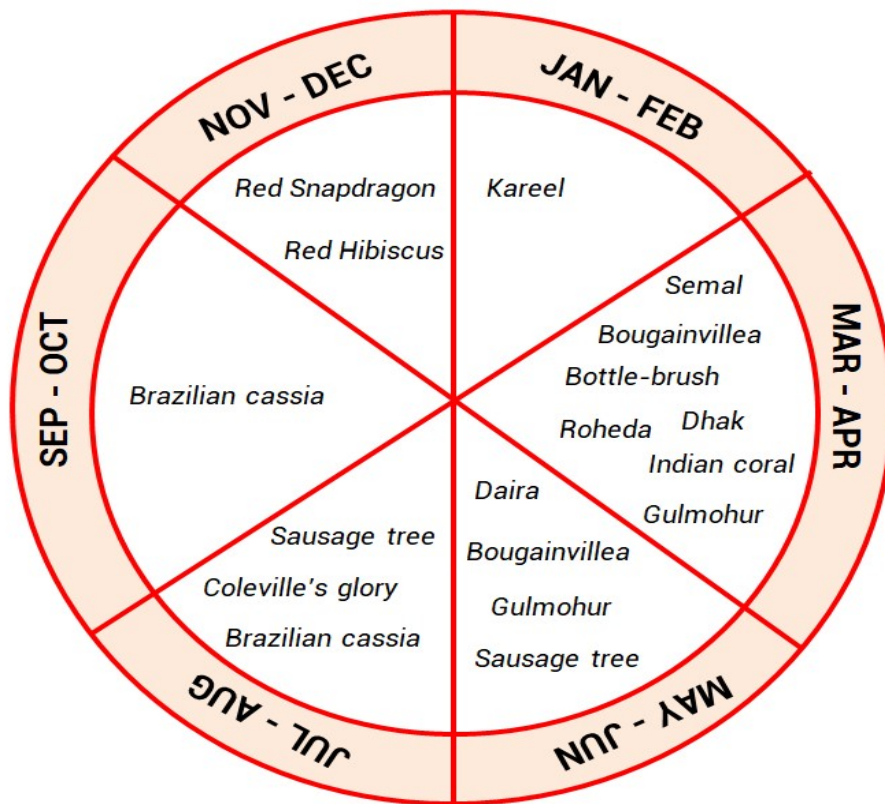
		areas, mound with height varying depending on water level).
4.	No. of plants per km	333
5.	Height of the saplings at the time of planting	1 .5 m to 2 m
6.	Survival percentage of plantation	90% at any time

Specifications for plants of last row

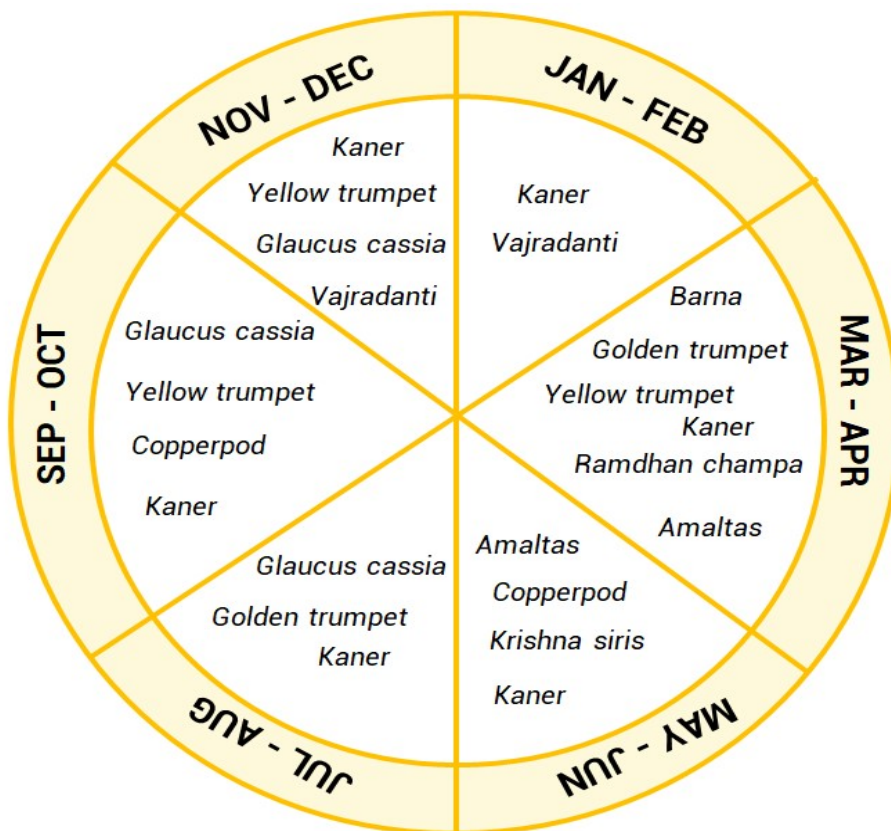
1.	Distance from preceding rows	3 m
2.	Spacing between plant to plant	8-12 m (6 m if high mortality expected)
3.	Size of the pits	60x60x60 cm (in alkaline soils, kankar panes to be broken by augur. In waterlogged areas, mound with height varying depending on water level).
4.	No. of plants per km	84 (167 at 6 m spacing)
5.	Height of the saplings at the time of planting	More than 2 m
6.	Survival percentage of plantation	90% after replacement of casualties in first two years. 80% afterwards.



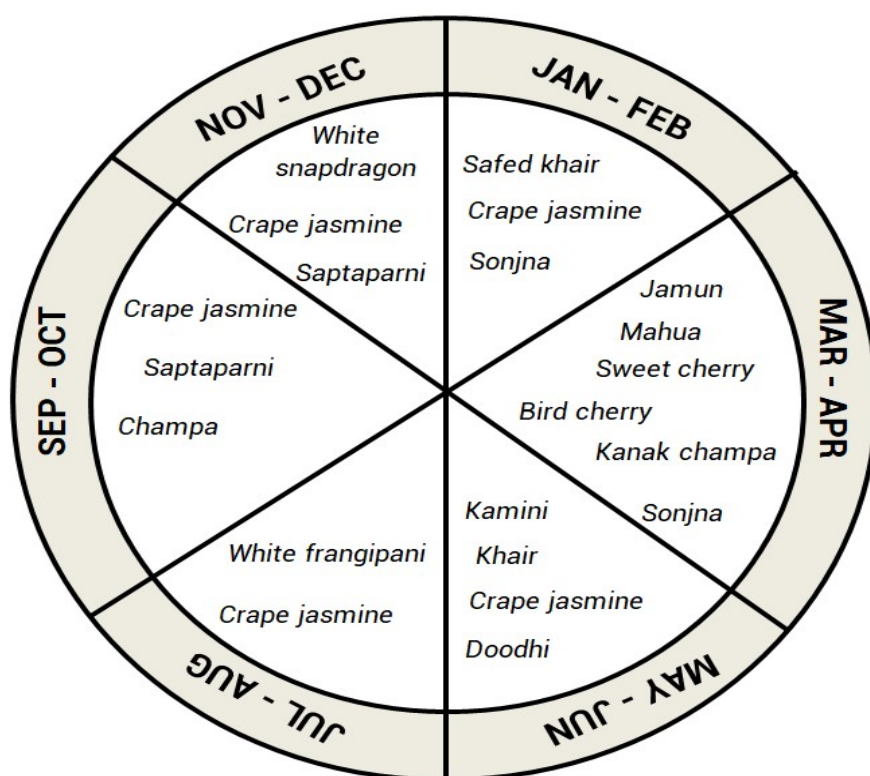
Flowering months of trees and shrubs with pink or purple flowers.



Flowering months of trees and shrubs with red or orange flowers



Flowering months of trees and shrubs with yellow flowers



Flowering months of trees and shrubs with white flowers

3.1.3.4 Initiatives By Departments/ Institutions (Govt./Pvt.)

All Departments/ Institutions (Government and Private) may adopt/ plant/ raise avenue plantations as prescribed along the roads adjoining their campuses.

3.1.3.5 Protection Measures

The fencing of single row plantations is generally done by using iron/brick/cement guards. Locally available bamboo guards or thorn fencing should be used where protection can be ensured through them.

For the grass turfing, the surface is to be prepared adequately for grass sowing. The grass and shrub planting is done to provide a strong surface cover but needs a well-prepared surface. All masses of loose debris should be removed. Any convexities should be removed and similarly any concavities are to be filled by good soil. The surface should have sufficient layer of good quality soil (upto 45 cm) so as to have better growth and survival of grasses and shrubs.

For more details regarding tree guards it is advised to refer the “Guidelines on Landscaping and Transplantation”, Indian Road Congress, 2009.

The Office Memorandum with indicative list of species for plantation is annexed herewith as **Annexure-III**.

3.1.3 Eco-restoration: Ecological restoration basically aims to assist an ecosystem move backwards in historical path to its near pristine form. While the restored ecosystem will not necessarily revert back its exact former state due to restrictions such as climatic change, urbanization, acclimatization of species etc., however, the general direction may be replicated. In order to restore an ecosystem a combination of knowledge of its former state, data from existing intact ecosystems (reference sites), and other ecological, cultural and historical information is needed. These sources will enable the identification of the historical trajectory, and aid in assisting the ecosystem towards health and integrity.

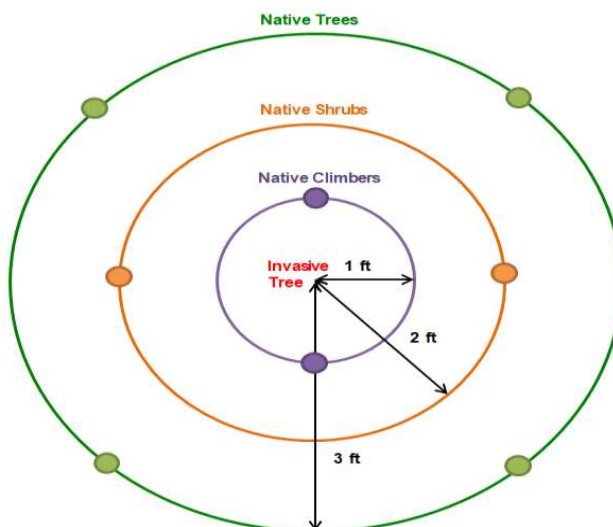
There are three main invasive alien species in the Delhi ridge eco-system are:

- a) *Prosopis juliflora* or Vilayati Kikar
- b) *Lantana camara*
- c) *Leucanea leucocephala* or Subabool

3.1.2.1 Standard Operating Protocol for Eco-restoration

Depending upon the type of Invasive Alien Species, different methodologies are prescribed for eco-restoration of any site.

- **Biological suppression through canopy lifting and planting of native species:**
 - i. **PRUNING** - The two major ways of pruning are thinning out (removing a branch to its origin) and heading back. Heading back is a method where the branch is cut back to a stub, small branch or bud. This method changes the shape of the tree or shrub noticeably. This way, the dense shoots emerges from just below the cut and are usually weakly attached and split out easily. It is one efficient and appropriate method to help in directing the growth of the trees (especially young trees).
 - ii. **PLANTING**- Invasive species like *Prosopis juliflora* compete with natives for resources. Thus, native species must be planted in a concentric circle in the following manner:



- The innermost circle at a distance of 1 foot from the Prosopis tree will consist of two native climbing species.
 - The middle circle at a distance of 2 feet from the tree will consist of two native shrub/understory species.
 - The outermost circle at a distance of 3 feet from the tree will consist of four native tree species.
- iii. WEEDING- In order to prevent re-emergence of young Prosopis saplings, weeding will be carried out twice a year. The pre-monsoon weeding will be conducted in June and the post-monsoon weeding will be conducted in October.

The restoration site will be maintained for a period of seven years. Within seven monsoon seasons, the natives will come to dominate the landscape.

In the case of *Leucanea leucocephala* / Subabool, the young sapling has to be removed manually on a regular basis and pruning of trees must be done before seed pods start to emerge on the tree. Elimination of Subabool is one extremely labour intensive and high maintenance process involving heavy costs but is probably the only effective method to control spread of this species.

- **Elimination of *Lantana camara* through the cut rootstock methodology**

Sites for removal of Lantana should be identified through field surveys using the data collected on the extent/ magnitude of infestation and density of Lantana. The Lantana removal operation should follow the 'inside-out' method wherein Lantana is removed first from areas with maximum Lantana density and then moving outwards along a decreasing Lantana density gradient. In areas having undulating terrain and hilly tracts Lantana removal operation should be taken up on the slopes first and then downwards to the valley. The removal operation should be done preferably at a time when a majority of the Lantana clumps are not in flowering or fruiting stage. The cut root stock method shall be employed for the removal of the Lantana.

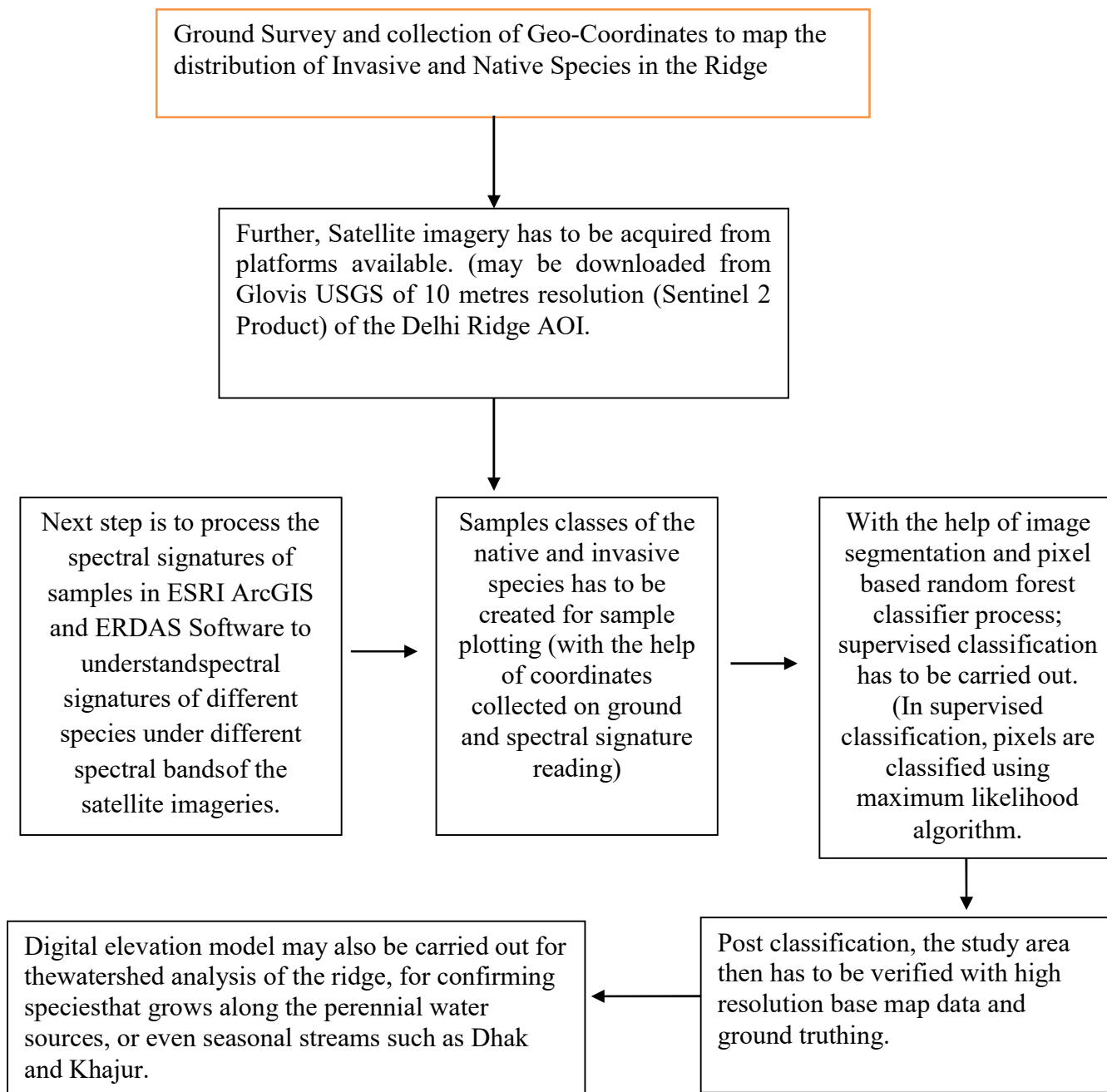
- **Strengthening of nurseries and availability of adequate funding**

Since for eco-restoration of Prosopis, the requirement is for 2+2+4 climbers, shrubs and trees per Prosopis to be suppressed adequate arrangement should be in place in the nursery also taking into consideration the casualty replacement and gap filling requirement as well. For Lantana area restoration there also needs to be in place adequate grass banks (Seeds/seedlings) in the nursery.

Adequate funding should also be in place for planning, implementation and monitoring phases of the restoration process.

- **Planning for eco-restoration**

Methodology followed to map the distribution of Invasive species:



Detailed SOP is annexed herewith as **Annexure-IV**.

3.1.4 High-density multilayered plantation: Afforestation of an area by planting of native species through high-density multilayered plantation.

3.1.5 Vertical Greening: Specialized type of urban garden that focuses on walls and roofs.

3.2 Development of online portals

3.2.1 Development of Online Portal for Green Action Plan

- Department will develop a dedicated portal for placing monthly progress of Green Action Plans along with geo locations of the plantations.
- Status of Third Party Audit of plantation sites having survived minimum of three seasons (i.e., 1.5 years from the date of plantation).
- Access to public about the details of the effort being putting by different Greening Agencies.

3.2.2 Creation of Online Portal for Distribution of Seedlings

- Department will develop a dedicated portal for facilitating distribution of seedlings to NGOs, individuals, Government Departments etc.
- The applicant will need to click a selfie with the seedling and to upload on the portal.
- Maintenance of real time data of distributed seedlings by the Department and ease of access to seedlings for general public.
- General public to be given access for viewing the efforts of Department.

3.3 Third Party Audit of Plantations

Forest Department to constitute a dedicated team with following objectives:

- To analyse and prepare database on the basis of third party monitoring of plantations taken up by Forest Department.
- To analyse third party audit reports received from the other Greening Agencies;
- To cross-check status of plantation sites of other Greening Agencies;
- To prepare database for site specific plant species on the basis of above findings.

3.4 Soil Moisture Conservation Works

The forests of Delhi also provide an important ecosystem service by recharging underground aquifers. Soil and moisture conservation is essential for river rejuvenation as it helps in enhancing percolation of water in the soil and lowering surface runoff and also keeping a check on soil erosion by ensuring retention of water in catchment area, improvement of

environmental flow and improvement of quality of water. This will eventually help in enhancing the site quality which ultimately improves the natural regeneration, plant survival and growth, forest productivity and biodiversity conservation. It improves the overall water quantity and quality.

Further, the Ministry of Environment, Forest and Climate Change, Government of India has prepared a Detailed Project Report on Rejuvenation of River Yamuna and the State Forest Departments have been made the Nodal Department for implementation of the same. As per the DPR, the model called YM/DL/CI/01: Rehabilitation of soil and water conservation structures has been suggested for the State of Delhi. The Soil Moisture Conservation Measures to be taken under the project are Conservation ditch, Contour trench, Contour bund, Loose boulder check dam, Brushwood check dam, Pond Earthen check dam, Gully plug etc.

The Department of Forests and Wildlife, GNCTD has identified areas around existing plantations for flood plain rejuvenation through creation of wetlands, biotic stabilization of banks with native vegetation, grassland development etc. These efforts are being made to consolidate the plantation area by including these susceptible patches of land and showcase the restoration of landscape in Yamuna Flood plain.

3.5 Outreach and Education

With institutionalization of sustainable development principles in development projects in Delhi, there has been demand for training the staff of development and enforcement agencies in extant acts and rules. Further, Hon'ble High Court has also directed to conduct training. Hence training for implementation of various acts like Indian Forest Act-1927, Wildlife (Protection) Act-1972, Forest Conservation Act-1980, Delhi Preservation of Trees Act-1994, National Biodiversity Act, 2005, etc. are being conducted by the Department for enforcement agencies, government agencies, NGOs, students, RWAs, general public.

4. Target for GAP 2023

During, 2023 the Govt. of NCT of Delhi has decided to plant around 20 lakh trees in addition to shrubs, bamboos and grasses in view of G-20 Summit, 2023.

The Green Action Plan has been divided into Summer and Winter Action Plan to control the pollution throughout the year and increase the Green Cover of the State.

The target for the year 2023 is as under:

Target for Green Action Plan 2023				
SN	Agency	Tree	Shrubs/Hedge	Total
Plantation				
1	Forest Department	12,00,000	5,00,000	17,00,000
2	DDA	3,00,000	5,50,000	8,50,000
3	MCD	2,00,000	3,24,965	5,24,965
4	NDMC	10,000	50,00,000	50,10,000
5	PWD	76,000	3,24,000	4,00,000
6	Education Department	1,00,000	1,75,000	2,75,000
7	Higher Education	1,00,000	0	1,00,000
8	NTPC	1,00,000	0	1,00,000
9	DSIIDC	19,440	1,80,560	2,00,000
10	DJB	920	99,080	1,00,000
11	BSES Rajdhani (BRPL)	17,930	12,070	30,000
12	CPWD	4,000	22,500	26,500
13	Northern Railways	0	20,000	20,000
14	DUSIB	500	12,000	12,500
15	Delhi Cantt.	150	10,000	10,150
16	NDPL	10,000	0	10,000
17	NHAI	10,000	50,000	60,000
18	DMRC	2,000	500	2,500
19	IFCD, GNCTD	1,926	68074	70,000
20	Department of Env. (Hort.)	1,500	0	1,500
21	DTC	780	495	1,275
Plantation Total		21,55,146	73,49,244	95,04,390
Free Distribution				
1	Forest	4,00,000		
2	DPGS	2,50,000		
3	CPWD	50,000		
Distribution Total		7,00,000		
Grand Total		1,02,04,390		

**GOVERNMENT OF NCT OF DELHI
DEPARTMENT OF FORESTS & WILDLIFE
A-BLOCK, 2nd FLOOR, VIKAS BHAWAN, I.P. ESTATE,
NEW DELHI-110002**

Suggestive list of native Ridge tree species for plantation

S. No.	Common Name	Botanical name
1.	Amaltas	<i>Cassia fistula</i>
2.	Amla	<i>Phyllanthus emblica</i>
3.	Amrood (Guava)	<i>Psidium</i>
4.	Arjun tree	<i>Terminalia arjuna</i>
5.	Bargad	<i>Ficus benghalensis</i>
6.	Balam Kheera	<i>Kigelia pinnata</i>
7.	Behada	<i>Terminalia bellirica</i>
8.	Belpatra	<i>Aegle marmelos</i>
9.	Ber	<i>Zizyphus mauritiana</i>
10.	Bishtendu	<i>Diospyros cordifolia</i>
11.	Chamrod	<i>Ehretia laevis</i>
12.	Desi babool	<i>Vachellia nilotica</i>
13.	Sahajan	<i>Moringa oleifera</i>
14.	Dhak	<i>Butea monosperma</i>
15.	Dhau	<i>Anogeissus pendula</i>
16.	Gamhar	<i>Gmelina arborea</i>
17.	Gular	<i>Ficus racemosa</i>
18.	Goya Khair	<i>Dichrostachys cinerea</i>
19.	Gulmohar	<i>Delonix regia</i>
20.	Jungle Jalebi	<i>Pithecellobium dulce</i>
21.	Hingot	<i>Balanites aegyptiaca</i>
22.	Imli	<i>Tamarindus indica</i>

Annexure-I

S. No.	Common Name	Botanical name
23.	Jhinjhiri	<i>Memecylon umbellatum</i>
24.	Karonda	<i>Carissa carandus</i>
25.	Khair	<i>Acacia senegal</i>
26.	Peepal	<i>Ficus religiosa</i>
27.	Phalsa	<i>Grewia asiatica</i>
28.	Phulai	<i>Senegalia modesta</i>
29.	Pilkhan	<i>Ficus virens</i>
30.	Putranjiva	<i>Drypetes roxburghii</i>
31.	Semal	<i>Bombax ceiba</i>
32.	Siras	<i>Albizia lebbek</i>
33.	Yellow flame	<i>Peltophorum pterocarpum</i>

Suggestive list of native Non-Ridge tree species for plantation

S. No.	Common Name	Botanical name
1.	Ashok	<i>Polyalthia pendula</i>
2.	Bakain	<i>Melia azederach</i>
3.	Harsingar	<i>Nyctanthes arbor-tristis</i>
4.	Jarul	<i>Lagerstroemia speciosa</i>
5.	India coral tree	<i>Erythrina variegata</i>
6.	Jamun	<i>Syzygium cumini</i>
7.	Harad	<i>Terminalia chebula</i>
8.	Kadam	<i>Neolamarckia cadamba</i>
9.	Kathal (Jack fruit)	<i>Artocarpus heterophyllus</i>
10.	Kachnar	<i>Bauhinia variegata</i>
11.	Kadi Patta	<i>Bergera koenigii</i>
12.	Kanak Champa	<i>Pterospermum acerifolium</i>

Annexure-I

S. No.	Common Name	Botanical name
13.	Lasora	<i>Cordia dichotoma</i>
14.	Maharukh	<i>Ailanthus excelsa</i>
15.	Mango	<i>Mangifera indica</i>
16.	Maulsari	<i>Mimusops elengi</i>
17.	Neem	<i>Azadirachta indica</i>
18.	Saptaparni	<i>Alstonia scholaris</i>
19.	Shahtoot	<i>Morus alba</i>
20.	Shisam	<i>Dalbergia sissoo</i>

**GOVERNMENT OF NCT OF DELHI
DEPARTMENT OF FORESTS & WILDLIFE
A-BLOCK, 2nd FLOOR, VIKAS BHAWAN, I.P. ESTATE,
NEW DELHI-110002**

Suggestive list of fruit tree species for plantation

S. No.	Common Name	Botanical name
1.	Amla	<i>Phyllanthus emblica</i>
2.	Amrood (Guava)	<i>Psidium</i>
3.	Akhrot / English Walnut	<i>Juglans regia</i>
4.	Almond	<i>Prunus dulcis</i>
5.	Anar	<i>Punica granatum</i>
6.	Ber	<i>Zizyphus mauritiana</i>
7.	Cherry	<i>Prunus avium</i>
8.	Imli	<i>Tamarindus indica</i>
9.	Jamun	<i>Syzygium cumini</i>
10.	Kathal (Jack fruit)	<i>Artocarpus heterophyllus</i>
11.	Khajur/ Date palm	<i>Phoenix dactylifera</i>
12.	Lemon	<i>Citrus limon</i>
13.	Mango	<i>Mangifera indica</i>
14.	Mosambi	<i>Citrus limetta</i>
15.	Orange	<i>Citrus sinensis</i>
16.	Phalsa	<i>Grewia asiatica</i>
17.	Peach	<i>Prunus persica</i>
18.	Shareepa	<i>Annona reticulata</i>

**GOVERNMENT OF NCT OF DELHI
DEPARTMENT OF FORESTS & WILDLIFE
A-BLOCK, 2nd FLOOR, VIKAS BHAWAN, I.P. ESTATE,
NEW DELHI-110002.**

F.8 (194)/Forest/HQ/Plg.-I/GAP2023-24/2022-23/54

Date: 27.04.2023

OFFICE MEMORANDUM

Guidelines for plantation in avenues, parks, gardens, city forests outside notified forest areas etc.

More than 20 Greening Agencies present in NCT of Delhi undertakes plantation drive under Green Action Plan across the State. However, it has been observed that avenue plantations are being done randomly without following any standard guidelines for selection of species of trees/shrubs/herbs.

Therefore, Department of Forests and Wildlife, GNCTD has prepared a Guideline for selection of species for avenue plantation (annexed herewith).

Hence, all the Greening Agencies, taking up plantation in avenues, parks, gardens, city forests outside notified forest areas etc. of NCT of Delhi, are requested to implement the Guideline annexed herewith and to ensure that minimum 25% seedlings of flowering species and 15% seedlings of fruit bearing species are planted in avenue plantation.

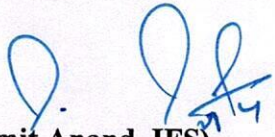
This issues with the approval of the Competent Authority.

(Enclosed as above)


(Amit Anand, IFS)
DCF (P&M)

Copy to:-

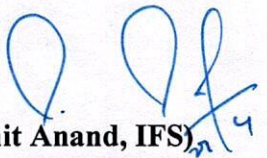
1. PS to Pr. Sec (E&F), Govt. of NCT of Delhi.
2. PS to PCCF, Department of Forests & Wildlife, Govt. of NCT of Delhi.
3. PA to CCF, Department of Forests & Wildlife, Govt. of NCT of Delhi
4. PA to CF, Department of Forests & Wildlife, Govt. of NCT of Delhi.


(Amit Anand, IFS)
DCF (P&M)

To,

1. The Vice Chairman (DDA), B Block, 1st Floor, Vikas Sadan, New Delhi-110073.
2. Pr. Commissioner (Personnel, Horticulture & Landscape), Delhi Development Authority, Vikas Sadan, INA, New Delhi-110023.
3. The Commissioner, MCD, Dr. S.P. Mukherjee Civic Center, JLN Marg, New Delhi-110002.
4. Chairman, New Delhi Municipal Council (NDMC), Palika Kendra, Parliament Street, New Delhi-110001.
5. Chairman, NHAI, National Highways Authority of India, G 5&6, Sector-10, Dwarka, New Delhi – 110075.
6. Chairman and Managing Director, NTPC Limited, NTPC Bhawan, SCOPE Complex, Institutional Area, Lodhi Road, New Delhi – 110003.
7. The Chief Executive Officer, Delhi Cantonment Board, Delhi-110010.
8. The Pr. Chief Engineer Northern Railways, Baroda House, Delhi/ the Divisional Engineer (Hort.), Office of the Northern Railway Divisional Office, State Entry Road New Delhi.
9. The Chief Engineer, (Gen), DMRC, Metro bhawan, Fire Brigade Lane, Barakhmbha Road, Delhi-110001.
10. The Manager (C), DTC, I.P. Estate, New Delhi-110002.
11. The CEO, Delhi Parks & Garden Society, 6th Floor, C-Wing, Delhi Secretariat, New Delhi.
12. The Superintending Engineer (Environment) DSIIDC, Technical Center Ground Floor Building Wazirpur Delhi-110052.
13. Director (HE), Directorate of Higher Education, BTE Complex, Muni Maya Ram Marg, Pitampura, Delhi-110034.
14. The Director, Education Department, Old Street, Delhi-110054.
15. The CEO/Director, DUSIB (Delhi Urban Shelter Improvement Board), ITO, New Delhi.
16. Director (Hort.), CPWD, A-Wing, Room No.- 101 Nirman Bhawan, New Delhi- 110011.
17. Commanding Officer, Eco Task Force, Bhatti Mines.
18. The Director (Hort.)/ Superintending Engineer, Horticulture Division, PWD, 13th Floor, MSO Building, I.P Estate, New Delhi – 110002.

19. CEO, Delhi Jal Boards, Addl. Chief Engineer (Plg) Drainage, 1st Floor Varunalaya Ph-I, Jhandewalan, Karol Bagh, New Delhi, Delhi 110005.
20. The Director (Hort.), Environment Department, 11th Floor, MSO Building, I.P. Estate, New Delhi 110002.
21. The Director (Hort.), Delhi Jal Board, Chandrawal Water Works No. 2, Civil Lines, Delhi-110054.
22. The Director (Hort), NDMC, Palika Kendra, New Delhi-110001.
23. The Director (Hort), Central Public Works Department (C.P. W.D) New Delhi, I.P. Estate.
24. Chief Engineer (I&FC), Zone-I, L.M. Bund Office Complex, Shastri Nagar, Delhi-110031.
25. Chief Engineer (I&FC), Zone-II, L.M. Bund Office Complex, Shastri Nagar, Delhi-110031.
26. The DGM (CSR), NDPL, Distt. Office, T-Block, Mangol Puri, Delhi-110083.
27. The CEO, BSES Rajdhani Power Ltd., BSES Bhawan, Nehru Place, New Delhi-110019.


(Amit Anand, IFS)
DCF (P&M)

PLANTATION MODEL FOR AVENUES, PARKS, GARDENS & CITY FORESTS OUTSIDE NOTIFIED FOREST AREA 2023



**Department of Forests and Wildlife,
Govt. of NCT of Delhi**

1. INTRODUCTION

Planting trees in the avenues, parks, gardens and city forests outside notified forest area of Delhi is of utmost importance due to the many benefits that plants provide. The capital has been grappling with severe air pollution, which has been identified as a major public health hazard. Trees have the ability to filter out pollutants from the air, making it cleaner and healthier to breathe. Additionally, trees absorb carbon dioxide and other greenhouse gases, helping to mitigate the effects of climate change. This is particularly relevant for Indian cities, which have been experiencing more extreme weather events such as heatwaves and flooding. Trees can also contribute to conserving water, reducing soil erosion, and creating habitats for wildlife, thus supporting biodiversity.

Planting flowering trees and shrubs is a simple but effective way of beautifying a city and improving the quality of life of its residents. Flowering trees add aesthetic value to the cityscape, making it visually appealing and attractive to visitors and residents alike. They provide a visual spectacle when in bloom, attracting visitors and residents to enjoy their stunning displays. This can have a positive impact on tourism, business, and even property values in the city.

Thus, the Department of Forests and Wildlife, Government of Delhi has prepared a scheme of plantation for the city's public agencies. The dual purpose for selecting this species is greening and beautification.

2. PLANTATION SCHEME

2.1 Categories

The planting of trees and shrubs is categorized in four categories:

- i. Tree plantation along roadways, turfing with grass and herbs;
- ii. Shrub plantation on medium strip, turfing with grass and herbs;
- iii. Block plantations in parks and gardens
- iv. Grid plantations in large patches of land

2.2 Selection of Species

Selection of trees, shrubs, herbs and grass species is based on following characteristics:

- (i) Location of plantation site
- (ii) Agro-climatic condition

- (iii) Ecological importance
- (iv) Presence, size and color of flowers
- (v) Value for pollinators
- (vi) Contribution towards pollution sequestration
- (vii) Growth rate
- (viii) Shape of the trunk
- (ix) Foliage and branching pattern
- (x) Shape and size of the canopy
- (xi) Properties of nitrogen fixation
- (xii) Soil-moisture conservation
- (xiii) Socioeconomic importance
- (xiv) Aftercare and maintenance requirement

A mix of showy species for beautification and local species commonly found in NCT of Delhi must be preferred.

Chart-1: Criteria for selection of species for roadside plantation¹

Agro-climatic Zone	Pollution Sequestration	Foliage Pattern	Growth Rate	Urban Beautification Value
As per the Agro-climatic map of ICAR, NCT of Delhi falls between the transition zone of upper and trans Gangetic Plains. (Figure-2)	(i) Canopy Shape and Size: - Dense foliage helps in better trapping of pollutants and also works as wind break. - Openness in canopy allows better filtration capacity. - Large canopy surface area works as better noise barrier. (ii) Leaf Area: - Large and broad leaf size accelerates better pollution retention. - Similarly leaves with complex shape and large circumference	Evergreen tree/shrub has higher pollution attenuation capacity as compared to deciduous or semi-deciduous tree/shrub.	Trees/shrubs with higher growth rate accelerate earlier canopy formation which is better for pollution sorption.	- Shape of canopy. - Presence, size, color and season of flowering. - Ability to provide nectar and nesting space pollinators.

¹ Sharma, Anuj. "Plantation Species Matrix for Roadside Plantation". Report for National Green Highways Mission-NHAI, 2016.

Agro-climatic Zone	Pollution Sequestration	Foliage Pattern	Growth Rate	Urban Beautification Value
	has higher dust capture capacity. (iii) Stomatal Index: High stomatal density, high index and large stomata give high transpiration rate thereby allowing higher sorption capacity. (iv) Tolerance: Selection of species with better tolerance capability for primary pollutants helps in maintaining the longevity of green corridor. (Figure-1)			

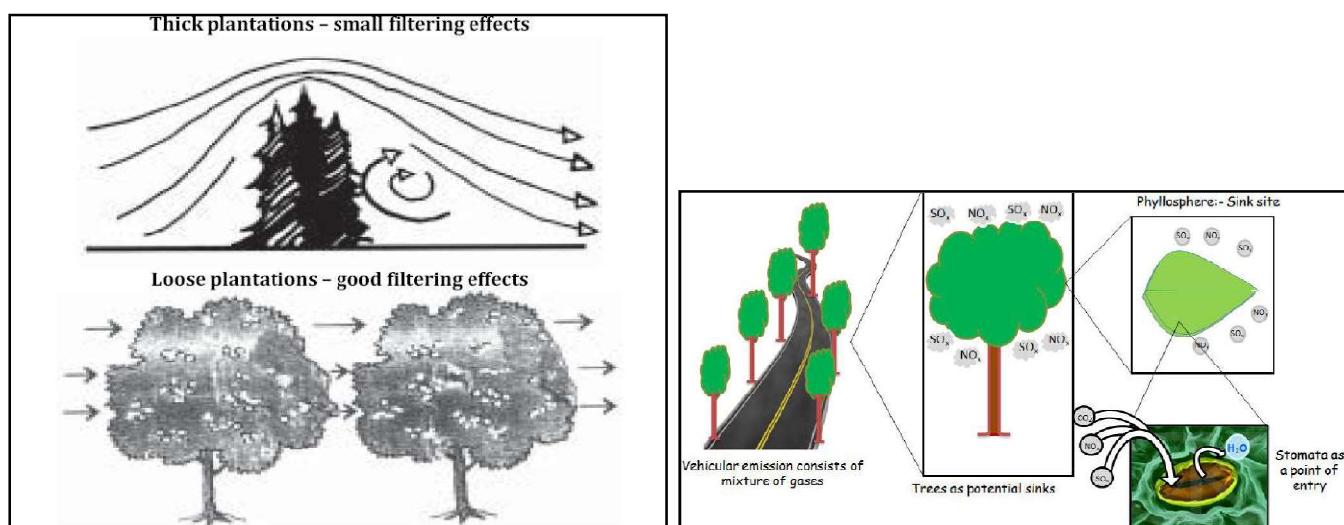


Figure-1: Pollution Sequestration by Roadside Plantation

(Source: Sharma, Anuj. "Plantation Species Matrix for Roadside Plantation". Report for National Green Highways Mission- NHAI, 2016)



Figure-2: Map of Agro-climatic Zones of India by Indian Council of Agricultural Research (ICAR)

2.3 Plantation Pattern: While finalizing the plantation pattern for any road side plantation, approach towards maintaining visual characteristics and uniformity in terms of landscape along the stretch to be considered. In cases of lacking land uniformity different schemes to be developed harmonizing the local variations in the design. To achieve this, entire stretch of the road corridor should be divided into homogenous landscape sections based on similarity in terms of agro-climatic conditions.

2.3.1 In accordance to the guidelines (2009) of Indian Road Congress on landscaping and transplantation the space available in the ROW is one major guiding factor for different themes of landscaping applied in the entire corridor. And as one of the main objectives of

roadside plantation is to provide shade, therefore, selection of plantation pattern for roadside plantation shall be designed as follows:

- (i) The first row along the Highways will be of small to medium sized ornamental trees.
- (ii) Subsequent rows, depending on the availability of width, will comprise of ornamental and/or shade bearing species, of more height than those in the first row.
- (iii) All these trees to be capable of attenuating pollution.
- (iv) Plantation of shrubs in the median.
- (v) Turfing with herbaceous species as ground cover in the median.
- (vi) Turfing with grass in the median.

3. INDICATIVE LIST OF SPECIES

Given below is an indicative list of species for plantation along avenues, roadsides, parks, forests, gardens, and city parks:

Table 2: Short-Medium Sized Trees Recommended for 1st Row of Roadside Plantation

S. No.	Botanical Name	Common Name	Flowering month	Flowering Color
1.	<i>Bauhinia purpurea</i>	Kaniar	October-December	Pink, lilac
2.	<i>Bauhinia variegata</i>	Kachnar	March	Pink, white
3.	<i>Bauhinia blakeana</i>	Hong Kong Orchid	January-March	Deep pink
4.	<i>Lagerstroma speciosa</i>	Jarul	Summer	Purple
5.	<i>Jacaranda mimosifolia</i>	Jacaranda	April-May	Purple-Blue
6.	<i>Melia azaderach</i>	Bakain	March to April	White-purple
7.	<i>Prunus dulcis</i>	Almond, badam	February-March	White-pink
8.	<i>Tabebuia impetiginosa</i>	Pink trumpet	February-March	Pink
9.	<i>Ochna obtusata</i>	Ramdhan Champa	Late April	Yellow
10.	<i>Tabebuia chrysotricha</i>	Golden trumpet	March-April, July-December	Yellow
11.	<i>Cassia fistula</i>	Amaltas	April-May	Yellow
12.	<i>Tecoma stans</i>	Yellow trumpet	March-October-November	Yellow
13.	<i>Calpurnia aurea</i>	Laburnum	March	Yellow
14.	<i>Koelreuteria elegans</i>	Flame-gold	Monsoon	Yellow
15.	<i>Peltophorum africanum</i>	African wattle	June-October	Yellow

S. No.	Botanical Name	Common Name	Flowering month	Flowering Color
16.	<i>Delonix regia</i>	Gulmohar	April-May	Red, yellow, orange
17.	<i>Colvillea racemosa</i>	Colville's Glory	July-August	Orange-red
18.	<i>Wrightia arborea</i>	Daira	May-June	Orange-red
19.	<i>Callistemon citrinus</i>	Bottle brush	February-March	Red
20.	<i>Erythrina blakei</i>	Blake's Coral Tree	March-April	Red
21.	<i>Psidium guajava</i>	Guava/ Amrud	February and June	White
22.	<i>Tabernaemontana divaricata</i>	Crape Jasmine	Almost throughout year	White
23.	<i>Dalbergia lanceolaria</i>	Takoli		White
24.	<i>Wrightia tinctoria</i>	Doodhi, sweet indrajau	May-June	White
25.	<i>Moringa oleifera</i>	Sonjna	February-March	White
26.	<i>Prunus avium</i>	Sweet Cherry	March to April	White
27.	<i>Prunus padus</i>	Bird Cherry	April	White
28.	<i>Senegalia catechu</i>	Khair	May to July	White

Table 3: Tall Tree Species Recommended for 2nd Row/3rd Row of Roadside Plantation

S. No.	Botanical Name	Common Name	Flowering month	Flowering Color
1.	<i>Albizzia lebbek</i>	Siris	April-May, June-July	Yellow-green
2.	<i>Albizia amara</i>	Krishna Siris	April-May	Yellow
3.	<i>Mangifera indica</i>	Mango	February-March	Yellow
4.	<i>Prosopis cineraria</i>	Khejri	March to May	Yellow
5.	<i>Neolamarckia kadamba</i>	Kadam	June to August	Yellow-orange
6.	<i>Schleichera oleosa</i>	Kosam	March	Yellow flowers, bright red new leaves
7.	<i>Crataeva adansonii</i>	Barna	April	White-Yellow-Purple
8.	<i>Tamarindus indica</i>	Tamarind/ Imli	June-August	Yellow-pink
9.	<i>Terminalia arjuna</i>	Arjun	April-May	Yellow-cream
10.	<i>Delonix regia</i>	Gulmohar	April-May/	Red, Orange, Yellow
11.	<i>Bombax ceiba</i>	Semal	March-April	Red, Orange, Yellow
12.	<i>Grevillea robusta</i>	Silky oak	March, Yellow	Orange
13.	<i>Butea monosperma</i>	Palash/ Dhak	March-April	Red-orange
14.	<i>Tecomella undulata</i>	Roheda	March	Red-orange
15.	<i>Pistacia chinensis</i>	Kakkar	March	Red
16.	<i>Kigelia africana</i>	Sausage Tree	April-August	Red
17.	<i>Melia azadirachta</i>	Bakain	March-April	Purple
18.	<i>Millettia peguensis</i>	Moulmein	March-April	Pink-Purple

S. No.	Botanical Name	Common Name	Flowering month	Flowering Color
		rosewood		
19.	<i>Lagerstroemia speciosa</i>	Jarul	April-June	Pink-Purple
20.	<i>Pterygota alata</i>	Buddha's coconut	March-May	Brown
21.	<i>Manilkara hexandra</i>	Khirni	January-February	White-cream
22.	<i>Madhuca longifolia</i>	Mahua	April	White cream
23.	<i>Azadirachta indica</i>	Neem	January to May	White
24.	<i>Hardwickia binata</i>	Anjan	August-September	White
25.	<i>Acacia ferruginea</i>	Safed khair	January-February	White
26.	<i>Syzygium cumini</i>	Jamun	April	White
27.	<i>Pterospermum acerifolium</i>	Kanak Champa	March-April	White (leaves similar to maple)
28.	<i>Alstonia scholaris</i>	Saptarni/ Devil Tree	October-December	White

Table 3: Tree Species Recommended for Median Plantation

S. No.	Botanical Name	Common Name	Flowering month/colour	Flowering Color
1.	<i>Erythrina variegata</i>	Indian coral tree	March-April	Red
2.	<i>Erythria suberosa</i>	Dhaul dhak	March-April	Red
3.	<i>Erythria stricta</i>	Coral	March-April	Red
4.	<i>Callistemon viminalis</i>	Bottle brush	February-March	Red
5.	<i>Cassia roxburgii</i>	Brazilian cassia	June to October	Red, orange pink
6.	<i>Ceiba speciosa</i>	Floss- silk tree	September –October	Pink
7.	<i>Gliricidia sepium</i>	Quickstick		Pink
8.	<i>Cassia javonia</i>	Java Cassia	April to June	Pink
9.	<i>Cassia religera</i>	Burmese pink cassia	May	Rose pink
10.	<i>Dichrostachys cinerea</i>	Sickle Bush, Shami, Goya Khair	June to September	Pink-yellow
11.	<i>Senna surattensis</i>	Glaucous cassia	June to December	Yellow
12.	<i>Senna siamea</i>	Kassod	June to December	Yellow
13.	<i>Grewia asiatica</i>	Phalsa	February-April	Yellow
14.	<i>Morus alba</i>	Shahtoot	February	Green
15.	<i>Plumeria obtusa</i>	White frangipani	April-May; August-September	White
16.	<i>Plumeria rubra</i>	Frangipani	May-July; August-September	White

17.	<i>Phyllanthus embelica</i>	Amla	February-May	White
18.	<i>Carissa carandas</i>	Karanda	March-April	White
19.	<i>Murraya paniculata</i>	Kamini	March-May	White

Table 4: Shrubs recommended for Median Plantation

S. No.	Botanical Name	Common Name	Type	Flowering month/colour	Flowering Color
1.	<i>Bougainvillea sp</i>	Bougainvillea	Shrub	April-August	Purple, orange, white, yellow
2.	<i>Tecoma stans</i>	Yellow trumpet/ Yellow Oleander	Shrub	March-October- November	Yellow
3.	<i>Thevetia nerifolia</i>	Kaner	Shrub	All year	Yellow
4.	<i>Parkinsonia aculeata</i>	Jerusalem thorn	Shrub	March-May	Yellow
5.	<i>Barleria prionitis</i>	Vajradanti	Shrub	November-January	Yellow
6.	<i>Calendula officinalis</i>	Calendula	Shrub/l arge herb	Autumn-winter	Yellow, Orange
7.	<i>Tabernaemontan adivaricata</i>	Crape Jasmine	Shrub	Almost throughout year	White
8.	<i>Antirrhinum</i>	Snapdragon	Large herb	Winter	White, Crimson, Pink
9.	<i>Callistephus chinensis</i>	China Aster	Shrub/ large herb		Purple, Pink
10.	<i>Nerium oleander</i>	Pink oleander	Shrub	April-August	Pink
11.	<i>Hibiscus rose-sinensis</i>	Shoe-flower	Shrub	All year	Red, Peach, Pink
12.	<i>Capparis decidua</i>	Kareel	Shrub	February-May	Red
13.	<i>Delphinium</i>	Larkspur	Shrub/ large herb	Winter	Blue

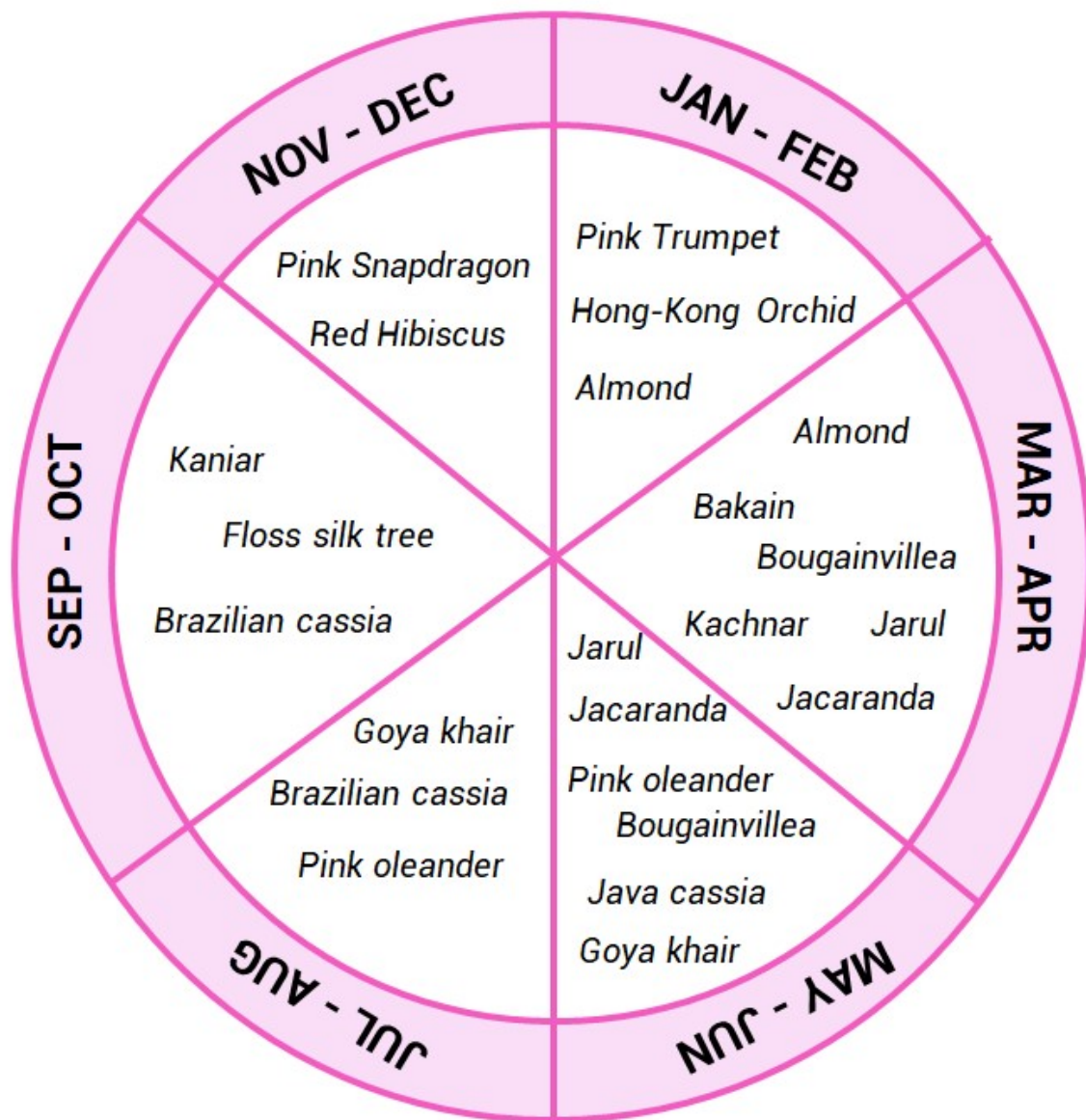


Figure 1: Flowering months of trees and shrubs with pink or purple flowers.

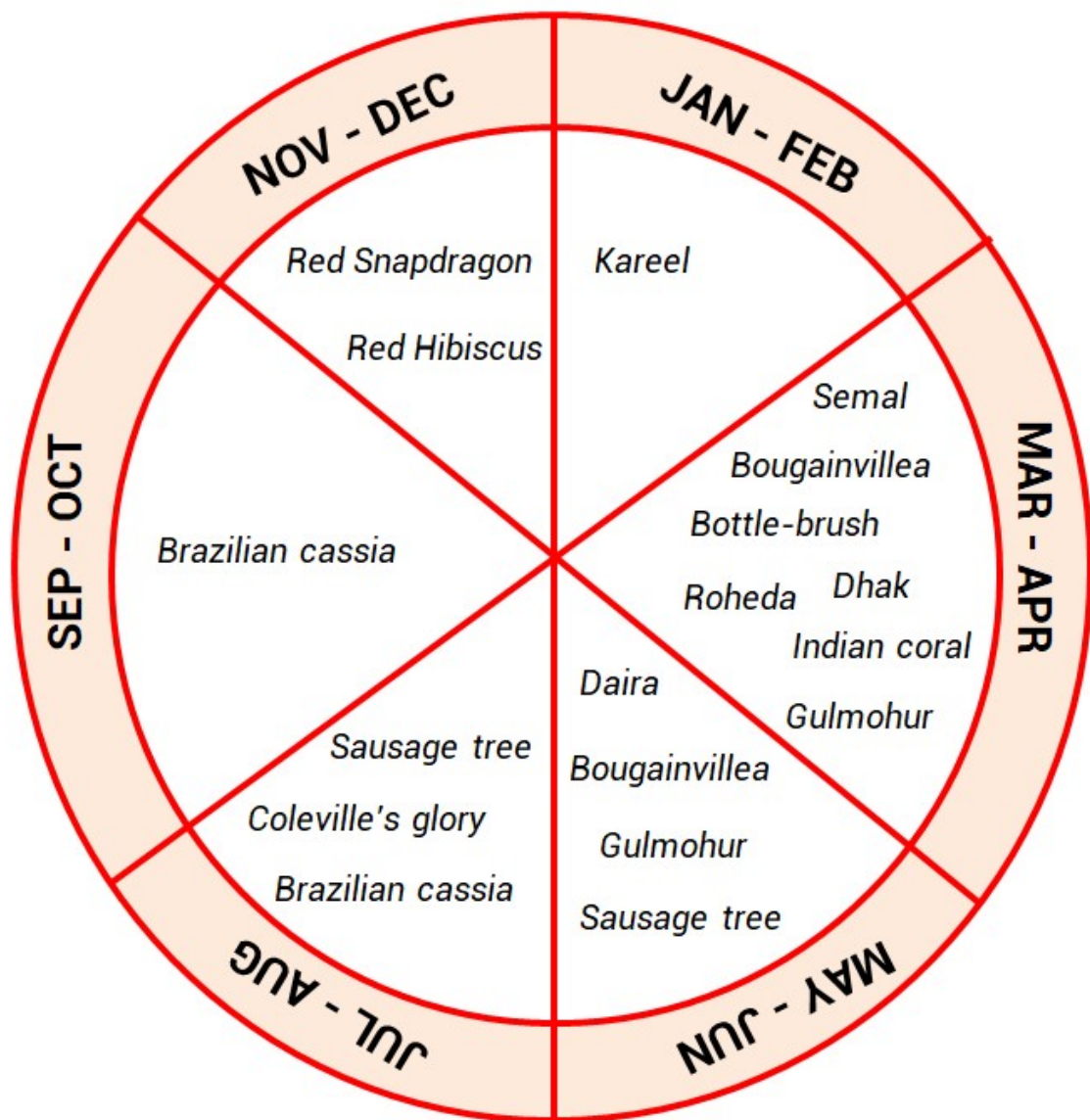


Figure 2: Flowering months of trees and shrubs with red or orange flowers.

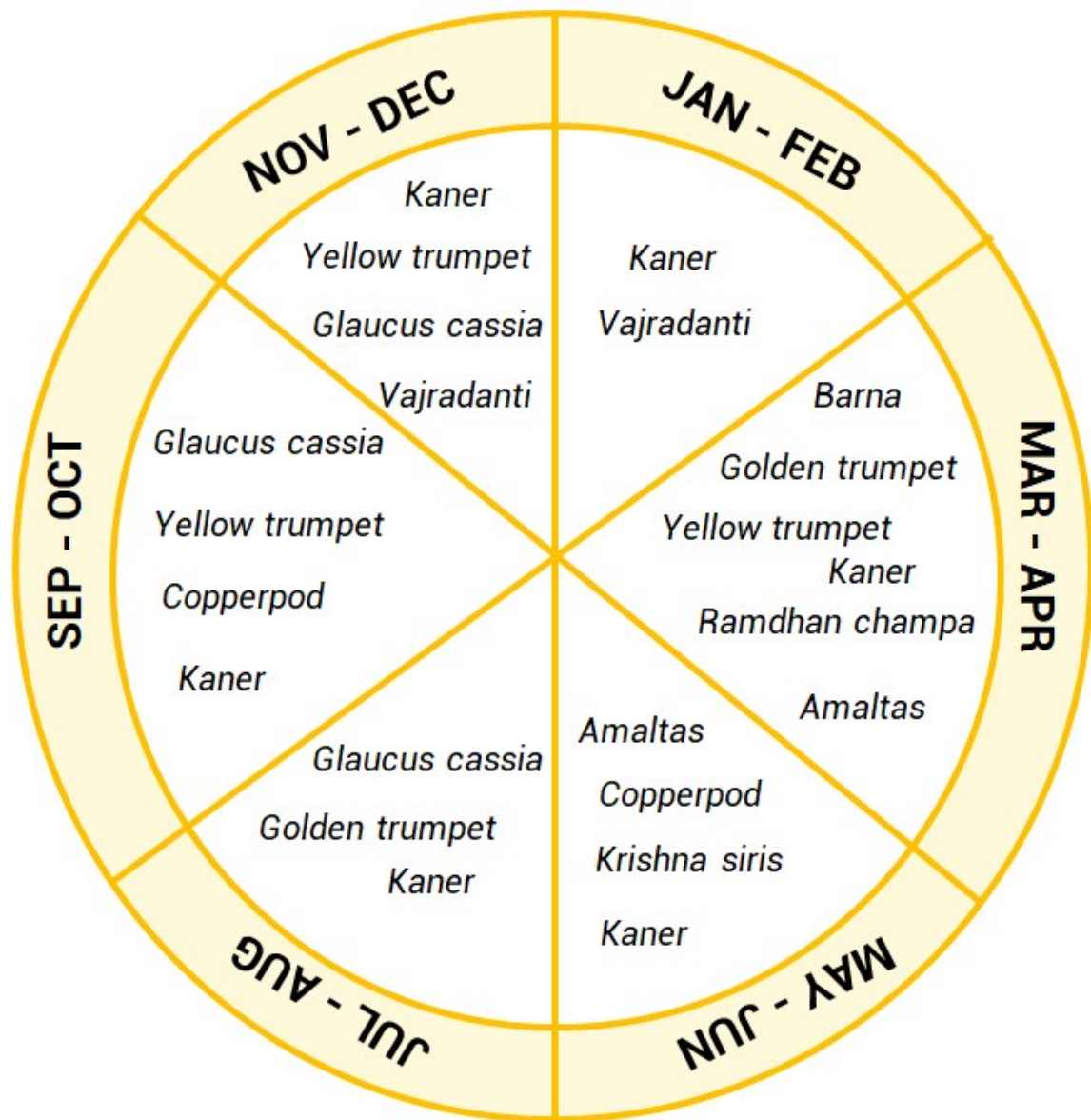


Figure 3: Flowering months of trees and shrubs with yellow flowers.

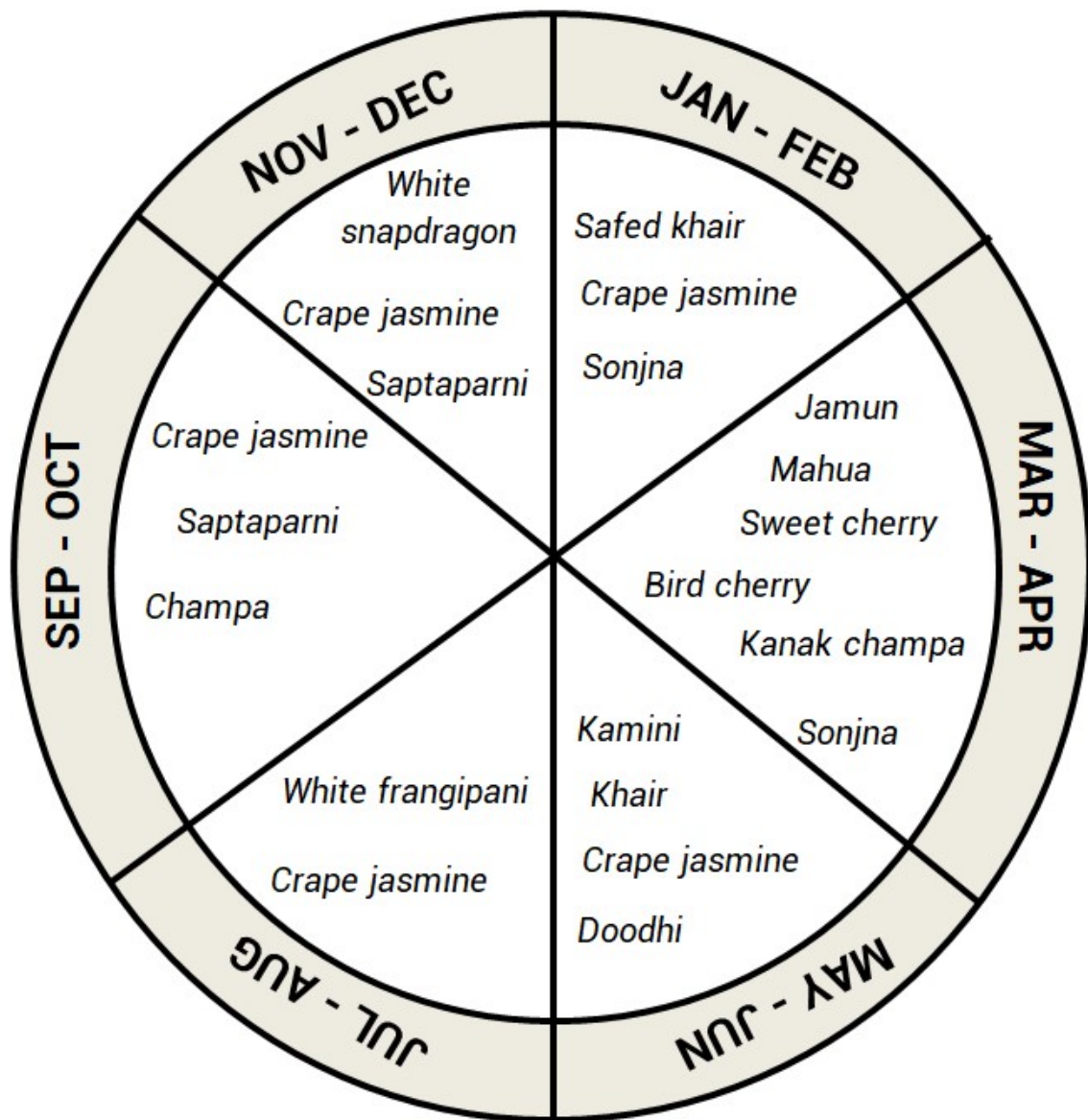


Figure 4: Flowering months of trees and shrubs with white flowers.

4. SPECIFICATIONS FOR ROAD PLANTATION

As per “Guidelines on Landscaping and Transplantation”, Indian Road Congress, 2009 the technical specifications for roadside plantation are given in Table 4 and Table 5 below:

Table 4: Specifications for ornamental plants (except last row)

1.	Spacing between plant to plant	3 m
2.	Spacing between rows	3 m
3.	Size of the pits	60x60x60 cm (in alkaline soils, kankar panes to be broken by augur. In waterlogged areas, mound with height varying depending on water level).
4.	No. of plants per km	333
5.	Height of the saplings at the time of planting	1.5 m to 2 m
6.	Survival percentage of plantation	90% at any time

Table 5: Specifications for plants of last row

1.	Distance from preceding rows	3 m
2.	Spacing between plant to plant	8-12 m (6 m if high mortality expected)
3.	Size of the pits	60x60x60 cm (in alkaline soils, kankar panes to be broken by augur. In waterlogged areas, mound with height varying depending on water level).
4.	No. of plants per km	84 (167 at 6 m spacing)
5.	Height of the saplings at the time of planting	More than 2 m
6.	Survival percentage of plantation	90% after replacement of casualties in first two years. 80% afterwards.

5. INITIATIVES BY DEPARTMENTS/ INSTITUTIONS (GOVT./PVT.)

All Departments/ Institutions (Government and Private) may adopt/ plant/ raise avenue plantations as prescribed along the roads adjoining their campuses.

6. PROTECTION MEASURES

The fencing of single row plantations is generally done by using iron/brick/cement guards. Locally available bamboo guards or thorn fencing should be used where protection can be ensured through them.

For the grass turfing, the surface is to be prepared adequately for grass sowing. The grass and shrub planting is done to provide a strong surface cover but needs a well-prepared surface. All masses of loose debris should be removed. Any convexities should be removed and similarly any concavities are to be filled by good soil. The surface should have sufficient layer of good quality soil (upto 45 cm) so as to have better growth and survival of grasses and shrubs.

For more details regarding tree guards it is advised to refer the “Guidelines on Landscaping and Transplantation”, Indian Road Congress, 2009.

I/72697/2023

**DEPARTMENT OF FORESTS & WILDLIFE
GOVERNMENT OF NCT OF DELHI
A-BLOCK, 2ND FLOOR, VIKAS BHAWAN
I.P.ESTATE, NEW DELHI-110002**

STANDARD OPERATING PROCEDURE (SOP)

FOR

ECOLOGICAL RESTORATION

OF

THE DELHI RIDGE ECOSYSTEM

CONTENTS

- 1. Goal of ecological restoration**
- 2. Invasive alien species in the Delhi ridge ecosystem**
- 3. Reference for ecological restoration**
- 4. Methodology**
 - A. *Prosopis Juliflora* (Vilayati Kikar)**
 - B. *Leucanea leucocephala* (Subabool)**
 - C. *Lantana Camara***
- 5. Planning for ecological restoration of Delhi ridge eco-system**

1. Ecological restoration and its goal

The Society for Ecological Restoration (SER)'s *International Primer on Ecological Restoration* (2004) defines ecological restoration as “**the process of managing or assisting the recovery of an ecosystem that has been degraded, damaged or destroyed** as a means of sustaining ecosystem resilience and conserving biodiversity.”

Ecological restoration basically aims to assist an ecosystem move backwards in its historical path to its near pristine form. While the restored ecosystem will not necessarily revert back to its exact former state due to restrictions such as climate change, urbanization, acclimatization of species etc., however, the general direction may be replicated. In order to restore an ecosystem a combination of knowledge of its former state, data from existing intact ecosystems (reference sites), and other ecological, cultural and historical information is needed. These sources will enable the identification of the historical trajectory, and aid in assisting the ecosystem towards health and integrity.

2. Invasive alien species in the Delhi ridge ecosystem

Biological invasion is the disruption in the functioning and processes of ecosystems and communities due to the increased abundance and distribution of a non-native species (Keane and Crawley, 2002). A species which causes such a disruption is called an invasive species. Invasive species typically exhibit one or more of the following characteristics:

- 1) Fast growth
- 2) Rapid reproduction
- 3) High dispersal ability
- 4) Phenotypic plasticity (the ability to alter growth form to suit current conditions)
- 5) Tolerance of a wide range of environmental conditions (Ecological competence)

There are three main invasive alien species in the Delhi ridge eco-system are

- a) *Prosopis juliflora* or Vilayati Kikar
- b) *Lantana camara*
- c) *Leucanea leucocephala* or Subabool

3. Reference ecosystem for eco-restoration of Delhi ridge

One of the key principles of ecological restoration is to have an appropriate local native reference ecosystem to guide project targets and serve as a model for planning a restoration project and later can be used as a basis for monitoring and evaluation purposes. The reference ecosystem may be an actual (physical) site or a conceptual model based on the data from different reference sites, historic records, field guides etc.

Mangar Bani is a sacred grove that lies on the boundary of Faridabad and Gurugram district of Haryana (~100 hectares), and **it is like a snapshot of what the northern Aravallis in Delhi would have looked like without urban expansion.** The Bani is covered by the beautiful Dhau (*Anogeissus pendula*) and Salai trees (*Boswellia serrata*), found on the slopes and top of the hills along with other species that are adapted to these dry, rocky hills. Using insights from field surveys in Mangar Bani, and data from literature such as *Flora of Delhi* (1963) by botanist J.K. Maheshwari, a publication on the native vegetation of Delhi was compiled by the Department of Forests and Wildlife, GNCT of Delhi. This publication is titled, **Native Plants of the Northern Aravalli Range in Delhi (2022)** and will serve as a reference for planting in this restoration project.

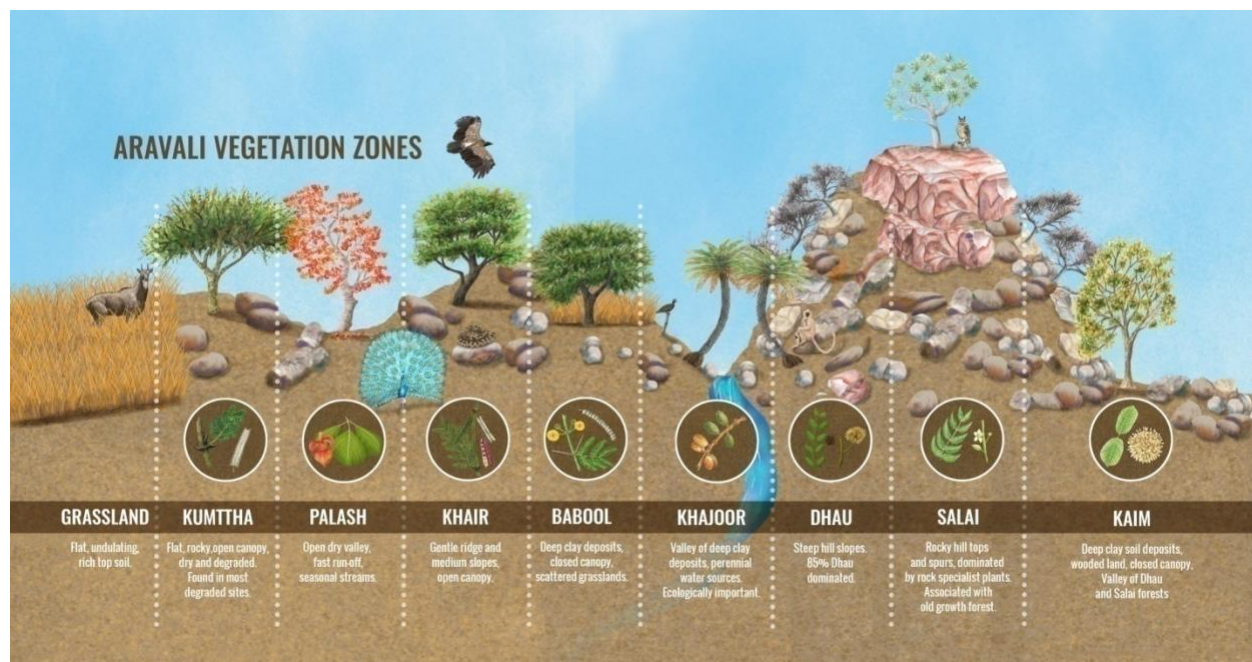


Figure 1: Microhabitats of the Northern Aravalli Range, Source: Richa Kedia, *Plants of the Northern Aravalli Range in Delhi*.

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Figure 1 describes different microhabitats of the Northern Aravalli Range. Microhabitat is a smaller part of the habitat which differs from the surrounding habitat. The physical conditions of a particular microhabitat can also be so different and unique to support unique species that may not be found in larger extent. These microhabitats also vary in temperature, humidity, exposure to light among many other factors.

The dry deciduous forest of the Delhi Ridge is an indicative of diverse microhabitats including Dhau (*Anogeissus pendula*), Kumpttha (*Senegalia senegal*), Palash (*Butea monosperma*), Kaim (*Mitragyna parviflora*), Khajur (*Phoenix sylvestris*), Desi babool (*Vachellia nilotica*), Khair (*Senegalia catechu*) and Salai (*Boswellia serrata*) forests. These species are specialized to the ridge habitat and are indicators of a particular kind of habitat and also an indicative of the topography of the ridge. These microhabitats are explained in detail in the book *Native Plants of the Northern Aravalli Range in Delhi*, published by the department in year 2022. To briefly put, starting with Dhau forests, these exist as pure forests and this micro-habitat is mostly found on steep rocky slope whereas Salai forests are found on top of the highest hills with ridges and spurs where soil is thin and dry and heavily dominated with big boulders. Kaim forests, on the other hand, found in valleys of the slopes on which Salai and Dhau forest stands. Kumpttha forest stands on gentler slope or flat rocky surface and they are often an indicative of degradation of the forests, with a wide open canopy. Khair forests are also found on the gentler ridges and slope of hills. Forest of Palash exists in drier open valleys or grows along the seasonal water flow, with deep soil deposits. Khajur forests are found in water abundant areas and in deep rocky valleys; areas either with water bodies or perennial streams. Desi Babool forests also occupy valleys with deep soil deposits, where water gets collected during monsoon.

To understand the type and physical aspects of the site and species composition, various habitats of the northern Aravalli range should be studied to include in the restoration plan.

4. Standard Operating Protocol for Eco-restoration.

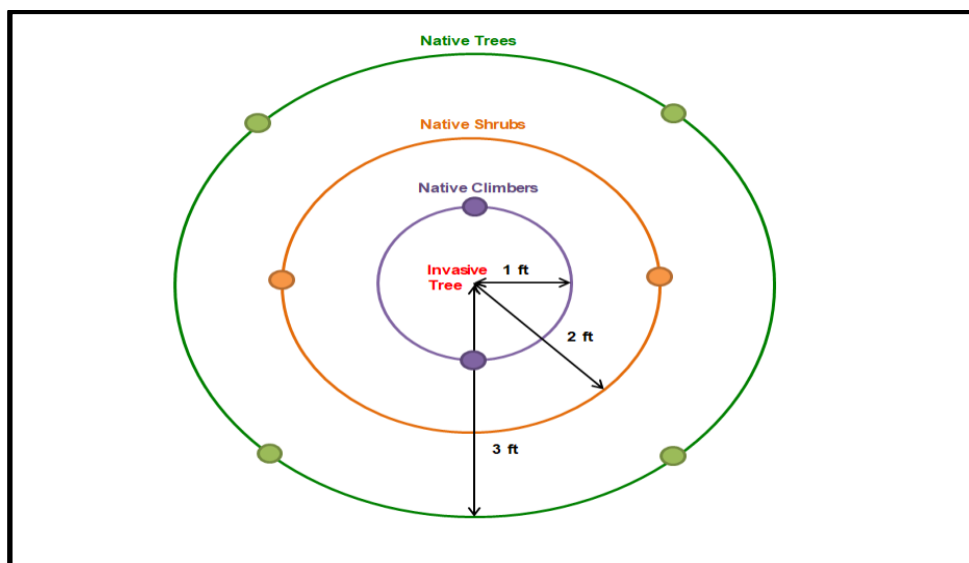
Depending upon the type of Invasive Alien Species, different methodologies are prescribed for eco-restoration of any site.

A. Biological suppression through canopy lifting and planting of native species

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Areas with *Prosopis juliflora* and *Leucaena leucocephala* (Subabool) are to be restored through **biological suppression through canopy lifting and planting of native species** is proposed. The standard operating protocol is detailed below:-

- i. **PRUNING** - The two major ways of pruning are thinning out (removing a branch to its origin) and heading back. Heading back is a method where the branch is cut back to a stub, small branch or bud. This method changes the shape of the tree or shrub noticeably. This way, the dense shoots emerges from just below the cut and are usually weakly attached and split out easily. It is one efficient and appropriate method to help in directing the growth of the trees (especially young trees).
- ii. **PLANTING**- Invasive species like *Prosopis juliflora* compete with natives for resources. Thus, native species must be planted in a concentric circle in the following manner



- The **innermost circle** at a distance of 1 foot from the *Prosopis* tree will consist of two native climbing species.
- The **middle circle** at a distance of 2 feet from the tree will consist of two native shrub/understory species.
- The **outermost circle** at a distance of 3 feet from the tree will consist of four native tree species.

- iii. WEEDING- In order to prevent re-emergence of young *Prosopis* saplings, weeding will be carried out twice a year. The pre-monsoon weeding will be conducted in June and the post-monsoon weeding will be conducted in October.

The restoration site will be maintained for a period of seven years. Within seven monsoon seasons, the natives will come to dominate the landscape.

In the case of *Leucanea leucocephala* / Subabool, the young sapling has to be removed manually on a regular basis and pruning of trees must be done before seed pods starts to emerge on the tree. Elimination of Subabool is one extremely labour intensive and high maintenance process involving heavy costs but is probably the only effective method to control spread of this species.

B. Elimination of *Lantana camara* through the cut rootstock methodology

Sites for removal of Lantana should be identified through field surveys using the data collected on the extent/ magnitude of infestation and density of Lantana. The Lantana removal operation should follow the ‘inside-out’ method wherein Lantana is removed first from areas with maximum Lantana density and then moving outwards along a decreasing Lantana density gradient. In areas having undulating terrain and hilly tracts Lantana removal operation should be taken up on the slopes first and then downwards to the valley. The removal operation should be done preferably at a time when a majority of the Lantana clumps are not in flowering or fruiting stage. The cut root stock method shall be employed for the removal of the lantana.

- The person, who engages in removal of Lantana, is positioned in a way that he stands near centre of the Lantana clump with his back facing the clump and holding the handle of digger (kudaal).
- Using the specially designed digger, the person cuts the main rootstock of Lantana 3–5 cm below the soil surface by hitting the rootstock 3 or 4 times; while hitting the rootstock the blade of the digger gets lodged into the main tap root, and at this point it is useful to move the handle of the digger in the forward direction away from the body of the person so as to sever the connection of the clump with the main tap root. In case the clumps of Lantana form impenetrable thickets, it is advantageous to cut the rootstocks of 3–4 contiguous clumps to make the removal operation convenient.

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- It may be noted that the branches of *Lantana* clumps should not be slashed/cut to gain access to the centre of the clump for its removal by cut rootstock method. The branches of *Lantana* thicket formed by more than one clump should be lifted and tipped over from one end by using a wooden or bamboo pole of about 1.5–2.5 m long and diameter 5–6 cm which is inserted just below the branches from one side and rolled over easily by two workers holding the pole at either end and pressing it so as to reach the centre of the clump. Such manual handling of impenetrable thicket is possible because of the umbrella type of canopy which makes it difficult to reach the centre of clump easily. Such physical maneuvers minimize or prevent regeneration from rooted cut branches when they fall on the ground.
- Lift the clump/s and place the clump/s upside down.
- If the clump is not placed upside down, the prostrate rooted branches and the aerial old branches having aerial roots at nodes may develop into adult plants when they come in contact with the soil. Therefore, the upside– down orientation of cut clumps is critical in the prevention of regeneration of *Lantana* from cut clumps. It may be noted that *Lantana* does not produce root suckers.
- After drying the clumps, the clumps may be used as fuel or burnt at the same site or all the dried clumps may be collected at one place and then burnt.

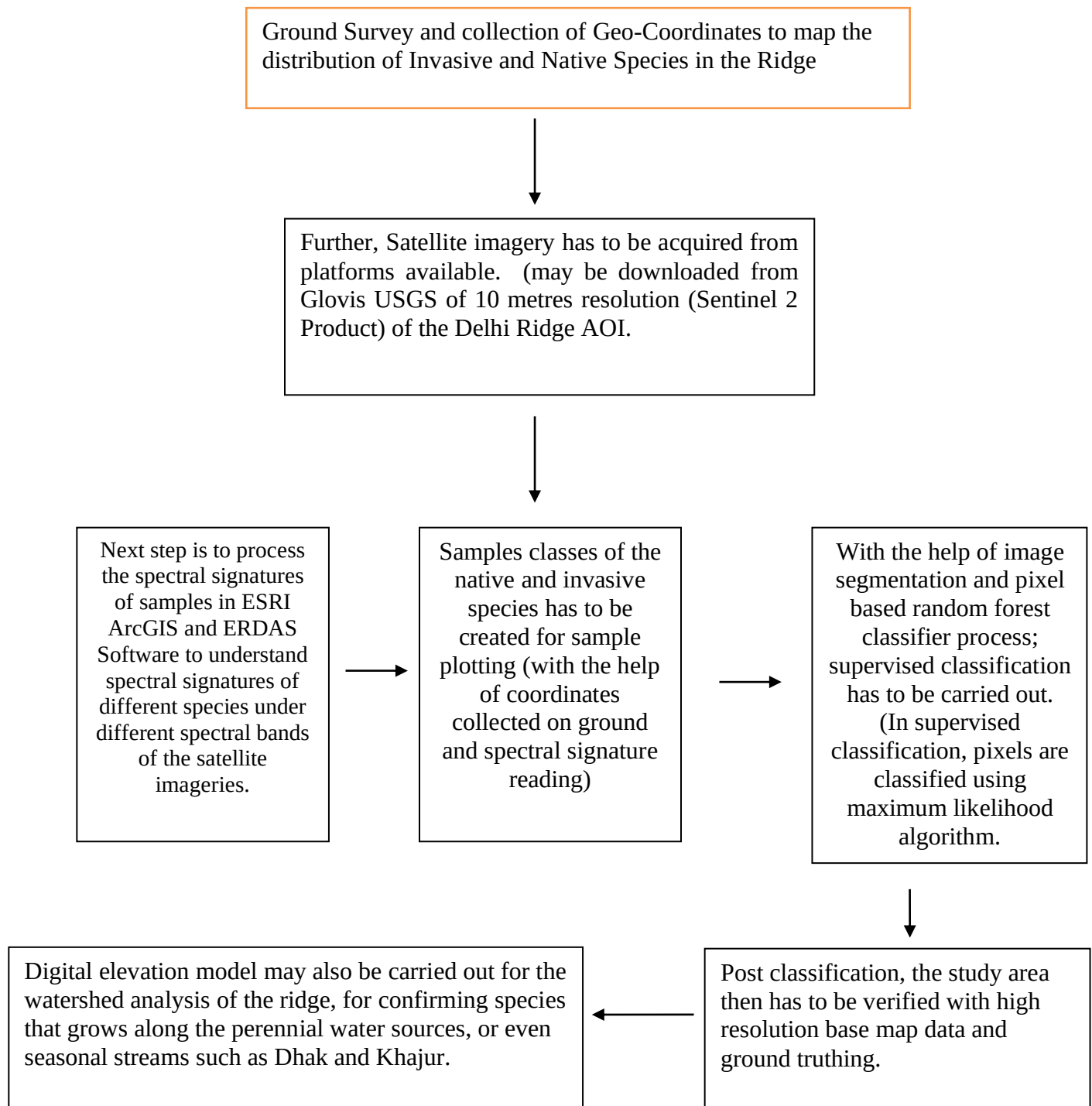
Continuous surveillance of the areas, where *Lantana* clumps have been removed and the areas beneath trees used for perching by birds from where the saplings are removed, is necessary consecutively for three growing seasons (monsoons) for the emergence of new saplings. Post removing the *lantana* clumps, broadcasting of seeds and plantation of native tree species has to be done immediately to avoid re-emergence of *Lantana*.

Weed-free landscapes in open areas can be easily restored to grassland communities by planting rooted ramets or clumps of native grass species or by broadcasting pellets containing seeds of grass species.

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4. Planning for eco-restoration

Methodology followed to map the distribution of Invasive species:



Result of classification then gives the extent of the distribution of invasive species.

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5. Strengthening of nurseries and availability of adequate funding

- Since for eco-restoration of *Prosopis*, the requirement is for 2+2+4 climbers, shrubs and trees per *Prosopis* to be suppressed adequate arrangement should be in place in the nursery also taking into consideration the casualty replacement and gap filling requirement as well. For Lantana area restoration there also needs to be in place adequate grass banks (Seeds/seedlings) in the nursery.
- Adequate funding should also be in place for planning, implementation and monitoring phases of the restoration process.

ANNEXURE –I

LIST OF NATIVE TREE SPECIES

S.No.	SPECIES	COMMON NAME	TYPE
1	<i>Aegle marmelos</i>	Bael	Tree
2	<i>Albizia amara</i>	Krishna siris	Tree
3	<i>Albizia lebbeck</i>	Siris	Tree
4	<i>Albizia odoratissima</i>	Kala siris	Tree
5	<i>Anogeissus pendula</i>	Dhau	Tree
6	<i>Balanites aegyptiaca</i>	Hingot	Tree
7	<i>Bauhinia racemosa</i>	Jhinjheri	Tree
8	<i>Butea monosperma</i>	Dhak	Tree
9	<i>Cassia fistula</i>	Amaltas	Tree
10	<i>Cordia dichotoma</i>	Lasoda	Tree
11	<i>Cordia gharaf</i>	Gondi	Tree
12	<i>Crateva religiosa</i>	Barna	Tree
13	<i>Dichrostachys cinerea</i>	Goya khair	Tree
14	<i>Diospyros cordifolia</i>	Bistendu	Tree
15	<i>Ehretia laevis</i>	Chamrod	Tree
16	<i>Ficus virens</i>	Pilkhan	Tree
17	<i>Flacourtia indica</i>	Bilangda	Tree
18	<i>Gmelina arborea</i>	Gamhar	Tree
19	<i>Holoptelea integrifolia</i>	Kanju/Chudail Papdi	Tree
20	<i>Mitragyna parvifolia</i>	Kadamb/Kaim	Tree
21	<i>Moringa concanensis</i>	Wild sonjna	Tree
22	<i>Moringa oleifera</i>	Sonjna	Tree
23	<i>Morus alba</i>	Shahtoot	Tree
24	<i>Phoenix sylvestris</i>	Khajoor	Tree
25	<i>Phyllanthus emblica</i>	Amla	Tree
26	<i>Prosopis cineraria</i>	Khejri	Tree
27	<i>Salvadora oleoides</i>	Khabbar	Tree
28	<i>Salvadora persica</i>	Peelu	Tree
29	<i>Schleichera oleosa</i>	Kusum	Tree
30	<i>Senegalia catechu</i>	Khair	Tree
31	<i>Senegalia modesta</i>	Phulai	Tree
32	<i>Senegalia senegal</i>	Kumththa	Tree
33	<i>Stercularia urens</i>	Kulu	Tree
34	<i>Syzygium cumini</i>	Jamun	Tree

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35	<i>Tamarindus indica</i>	Imli	Tree
36	<i>Tecomella undulata</i>	Roheda	Tree
37	<i>Terminalia bellirica</i>	Baheda	Tree
38	<i>Vachellia leucophloea</i>	Ronjh	Tree
39	<i>Vachellia nilotica</i>	Desi babool	Tree
40	<i>Wrightia arborea</i>	Kala indrajao	Tree
41	<i>Wrightia tinctoria</i>	Doodhi	Tree
42	<i>Ziziphus mauritiana</i>	Ber	Tree

ANNEXURE –II

LIST OF NATIVE SHRUBS

S.No.	SPECIES	COMMON NAME	TYPE
1	<i>Abutilon indicum</i>	Kanghi/Indian mallow	Shrub
2	<i>Barleria prionitis</i>	Vajradanti	Shrub
3	<i>Calotropis gigantea</i>	Safed aak	Shrub
4	<i>Calotropis procera</i>	Aak	Shrub
5	<i>Capparis decidua</i>	Kareel	Shrub
6	<i>Capparis sepiaria</i>	Heens	Shrub
7	<i>Carissa spinarum</i>	Jungli karaunda	Shrub
8	<i>Ficus palmata</i>	Anjeeri	Shrub
9	<i>Grewia asiatica</i>	Falsa	Shrub
10	<i>Grewia flavescens</i>	Pisangna	Shrub
11	<i>Grewia tenax</i>	Gangeti	Shrub
12	<i>Gymnosporia senegalensis</i>	Kankera	Shrub
13	<i>Justicia adhatoda</i>	Adusa	Shrub
14	<i>Lawsonia inermis</i>	Mehendi	Shrub
15	<i>Murraya koenigii</i>	Curry patta	Shrub
16	<i>Nyctanthes arbor-tristis</i>	Harshingar	Shrub
17	<i>Tamarix dioica</i>	Jhau	Shrub
18	<i>Withania somnifera</i>	Ashwagandha	Shrub
19	<i>Ziziphus nummularia</i>	Jhad ber	Shrub
20	<i>Ziziphus oenoplia</i>	Makora	Shrub

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ANNEXURE –III

LIST OF NATIVE CLIMBERS

S.No.	SPECIES	COMMON NAME	TYPE
1	<i>Clitoria ternatea</i>	Blue pea	Climber
2	<i>Trichosanthes cucumerina</i>	Jungle chanchinda	Climber
3	<i>Combretum indicum</i>	Madhumalti	Climber
4	<i>Coccinia grandis</i>	Tindora/Ivy Gourd	Climber
5	<i>Cardiospermum halicacabum</i>	Kanphuta/Balloon vine	Climber
6	<i>Vallaris solanacea</i>	Roth/Bread flower	Climber
7	<i>Tylophora indica</i>	Dam bel	Climber
8	<i>Cocculus hirsutus</i>	Patalgarudi/broom creeper	Climber
9	<i>Cissampelos pareira</i>	Patha	Climber
10	<i>Dregea volubilis</i>	Hemajivanti	Climber
11	<i>Gmelina asiatica</i>	Badhara	Climber
12	<i>Tinospora cordifolia</i>	Giloy	Climber
13	<i>Abrus precatorius</i>	Ratti	Climber

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ANNEXURE –IV

LIST OF NATIVE HERBS

S.No.	SPECIES	COMMON NAME	TYPE
1	<i>Asparagus racemosus</i>	Satavari	Herb
2	<i>Ruellia prostata</i>	Neelambaram	Herb
3	<i>Chenopodium murale</i>	Khartua	Herb
4	<i>Blepharis maderaspatensis</i>	Doodhiya choti	Herb
5	<i>Tridax sp.</i>	Bhringraj	Herb
6	<i>Eclipta prostata</i>	Bhringraj	Herb
7	<i>Malvastrum coromandelianum</i>	Three lobed false mallow	Herb
8	<i>Sida acuta</i>		Herb
9	<i>Oxalis corniculata</i>	Common woodsorrel	Herb
10	<i>Oxalis debilis</i>	Pink woodsorrel	Herb
11	<i>Oxalis latifolia</i>	Large leaved sorrel	Herb
12	<i>Stellaria media</i>	Chickweed	Herb
13	<i>Spergula arvensis</i>		Herb
14	<i>Commelina sp.</i>	Dayflower	Herb
15	<i>Euphorbia helioscopia</i>	umbrella milkweed	Herb
16	<i>Indigofera linnaei</i>	nine-leaved indigo	Herb
17	<i>Melilotus indicus</i>	Jungli methi	Herb
18	<i>Fumaria parviflora</i>	Indian fumitory	Herb
19	<i>Portulaca pilosa</i>	pink purslane	Herb
20	<i>Anagallis arvensis</i>		Herb
21	<i>Convolvulaceae nummularius</i>	round leaved binweed	Herb
22	<i>Rumex spinosus</i>	Kandiali Palak	Herb